

Estimating the Causal Effect of Higher Education on Antenatal Care (ANC) Utilization with Sensitivity Analysis Using Machine Learning Methods

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1 Background of the study

Maternal health remains a major public health concern in low- and middle-income countries (LMICs), particularly in regions such as South Asia and Sub-Saharan Africa. Antenatal care (ANC) is a crucial component of maternal healthcare, providing opportunities to monitor the health of pregnant women and their unborn children, identify and manage pregnancy-related complications, and deliver essential preventive and health-promoting interventions. Adequate utilization of ANC services is therefore important for improving maternal and child health outcomes. However, despite substantial progress in maternal healthcare, inequalities in ANC utilization persist, with maternal education, socioeconomic status, place of residence, and other demographic factors playing important roles ([World Health Organization, 2016](#); [Islam and Masud, 2018](#); [Aktar and Sultana, 2026](#)). Adequate utilization of antenatal care (ANC) is essential for improving maternal and newborn health outcomes. ANC enables the early detection and management of pregnancy-related complications and provides important preventive, nutritional, and health education services. Inadequate ANC utilization may increase the risk of adverse maternal and neonatal outcomes, including complications during pregnancy, preterm birth, and low birth weight ([World Health Organization, 2016](#); [Islam and Masud, 2018](#)).

Bangladesh has made substantial progress in antenatal care (ANC) utilization over recent decades. However, disparities in ANC utilization persist across socioeconomic and educational groups. Ensuring equitable access to adequate ANC remains important for improving maternal and newborn health and achieving Sustainable Development Goal 3 (SDG 3) ([BDHS, 2022](#); [United Nations, 2015](#)).

Previous studies in Bangladesh have identified maternal education, socioeconomic status, place of residence, and other demographic factors as important determinants of ANC utilization ([Islam and Masud, 2018](#); [BDHS, 2022](#)). Higher maternal education is generally associated with greater ANC utilization. However, these findings are largely associational and do not establish a causal relationship between higher education and ANC utilization. Therefore, causal inference methods are needed to estimate the effect of higher education while accounting for potential confounding factors.

In this study, we aim to estimate the causal effect of higher education on Antenatal Care (ANC) utilization in Bangladesh, using data from the Bangladesh Demographic and Health Survey 2022 (BDHS 2022). Causal inference for socio-demographic data is challenging due to the non-experimental nature of the data. There may be the presence of confounding in these type of data which are difficult to handle ([Athey and Imbens, 2016](#)). Technical features and affordances may introduce selection bias that can affect the validity of causal inference. Randomized experiment is the ideal

solution but expensive in terms of time, money and effort. So, causal inference for non-experimental or observational data is of interest to researchers.

Methods such as propensity score matching, inverse probability weighting, and stratification are commonly applied to observational data to estimate causal effects ([Rosenbaum and Rubin, 1983](#)). Traditional causal methods assume a pre-specified functional form and often rely on the assumptions of linearity and additivity in the relationships between outcome and covariates. Therefore, these methods may fail to capture nonlinearities, complex interactions, and other relationships among covariates, treatments, and outcomes that cannot be specified a priori, potentially leading to biased or oversimplified estimates of causal effects ([Hill, 2011](#)).

Machine learning (ML) approaches for causal inference, including causal forests, Bayesian additive regression trees, and causal boosting, provide a flexible alternative to traditional causal models. These models do not require a pre-specified functional form or any prior assumptions about the relationships between outcome variable, treatment and covariates. Machine learning algorithms can effectively model high-dimensional and nonlinear relationships in the data and enable the estimation of heterogeneous treatment effects, in which the impact of a treatment differs across individuals or subgroups ([Wager and Athey, 2018](#)). By relaxing assumptions and enabling the data to capture nonlinear relationships, high-dimensional interactions, and other complex patterns, machine learning methods overcome the limitations of traditional approaches, enhancing the accuracy and robustness of causal effect estimates.

In the context of maternal early marriage and stunting in Bangladesh, socio-demographic, geographic, health-related, maternal, and parental factors interact in complex and multifaceted ways. These complex interactions and relationships may not be adequately captured by traditional causal methods. But, Machine learning-based causal inference methods may enable the modeling of complex relationships and interactions between maternal early marriage and child stunting. Consequently, ML-based causal inference methods can provide policymakers with more precise evidence to design effective policies aimed at reducing child stunting.

Even after adjusting for many observed covariates, causal estimates from observational data may remain vulnerable to unmeasured confounding ([Kasza et al., 2017](#)). In the context of maternal early marriage and child stunting in Bangladesh, key factors such as cultural norms, maternal autonomy, household decision-making, intra-household resource allocation, and community-level influences are often difficult to capture in large-scale survey data ([Rahman et al., 2015](#); [Holland and Rammohan, 2019](#)). If these unmeasured factors jointly influence maternal early marriage and child stunting, the estimated causal effect may be biased.

Even though machine learning-based causal inference approaches can capture complex, nonlinear,

and high-dimensional relationships among observed variables, they still rely on the assumption that there are balances with respect to unmeasured confounders, an assumption that cannot be verified in observational studies (Pearl, 2009). Because early marriage and child nutrition are shaped by complex socio-cultural factors, the assumption of no unmeasured confounding is especially stringent in this context, and any violation of this assumption could undermine the validity of the causal inferences drawn.

Therefore, sensitivity analysis is essential to evaluate the robustness of the estimated causal effect to potential unmeasured confounding. Sensitivity analysis formally evaluates how strong an unobserved confounder would need to be to substantially alter the estimated causal effect, thereby enhancing the transparency and credibility of causal conclusions (Cinelli and Hazlett, 2020; VanderWeele and Ding, 2017).

2 Statement of the problem

Antenatal care (ANC) utilization remains an important public health issue in Bangladesh despite substantial progress in maternal healthcare. Although previous studies have consistently reported a positive association between maternal education and ANC utilization, most rely on conventional regression methods and observational data, which primarily establish associations rather than causal relationships.

Maternal education is closely related to socioeconomic status, place of residence, household characteristics, and other factors that may also influence ANC utilization. Therefore, the observed association between higher education and ANC utilization may be affected by confounding. Moreover, conventional parametric models may be limited in capturing complex, nonlinear, and heterogeneous relationships among variables. Even when advanced causal methods are applied to observational data, estimates may remain sensitive to unmeasured confounding.

Therefore, this study aims to estimate the causal effect of higher education on ANC utilization using observational data from Bangladesh, employing machine learning-based causal inference methods to account for complex relationships and sensitivity analysis to assess the robustness of the estimated causal effect to potential unmeasured confounding.

3 Objectives of the Study

The primary objective of this study is to estimate the average causal effect of higher education on ANC utilization in a systematic way using observational data. More specific objectives are:

- Providing a comprehensive review of contemporary machine learning and Bayesian approaches

for estimating causal effects of a binary treatment on a binary outcome.

- Making a systematic and thorough comparison of these machine learning and Bayesian methods with classical parametric regression using a simulation study based on BDHS 2022 database.
- Suggesting a reliable method for estimating the causal effect of maternal early marriage on child stunting.
- Assessing the potential bias from unmeasured confounders on causal effect estimates for the selected method and evaluate the robustness of the findings through sensitivity analysis.

4 Methodology

The conditional average treatment effect (CATE) is an effective estimand for capturing heterogeneity in treatment effects. Conditional mean regression methods are well suited for estimating CATEs, as they can identify regions of the covariate space where the variability of the outcome parameter is relatively low. We investigate two main categories of methods for estimating conditional mean outcomes. The first category includes G-computation approaches, such as Bayesian Adaptive Regression Trees (BART). The second category consists of causal learning techniques, including Neural Networks, Causal Trees, Causal Forests, and Causal Boosting . Inverse probability weighting (IPW) is employed as a benchmark method to estimate causal effects.

The conditional average treatment effect (CATE) is an effective estimand to assess Heterogeneous Treatment Effects (HTE). Within the potential outcomes framework (Holland, 1986; Rubin, 1974), CATE can be formally defined as follows. Consider a sample of n individuals indexed by $i = 1, \dots, n$. For each individual i , let X_i denote the vector of baseline covariates, $Z_i \in \{0, 1\}$ the binary treatment indicator, and $Y_i \in \{0, 1\}$ the binary outcome. Also consider the two potential outcomes for individual i : $Y_i(1)$ under treatment ($Z_i = 1$) and $Y_i(0)$ under control ($Z_i = 0$). The individual treatment effect is defined as

$$\tau_i = Y_i(1) - Y_i(0).$$

But only one of the two potential outcomes is observable for any given individual. Thus, the observed outcome is

$$Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0).$$

Consequently, we cannot observe the treatment effect at the individual level, τ_i . This is known as the fundamental problem in causal inference. Within this framework, we assume conditional

exchangeability, $\{Y(1), Y(0)\} \perp\!\!\!\perp Z \mid X$, and positivity, $0 < P[Z = 1 \mid X = x] < 1$. The term $P[Z = 1 \mid X = x]$ represents the propensity score, $e(x)$, which is the probability of receiving the treatment given the baseline covariates. Under these assumptions, the conditional mean functions can be defined as:

$$\mu_1(x) = \mathbb{E}[Y_i \mid X_i = x, Z_i = 1], \quad \mu_0(x) = \mathbb{E}[Y_i \mid X_i = x, Z_i = 0].$$

The conditional average treatment effect (CATE) can then be written as:

$$\tau(x) = \mu_1(x) - \mu_0(x).$$

The average treatment effect (ATE), $\tau = \mathbb{E}[Y_i^1 - Y_i^0]$, can be obtained by taking the expectation of $\tau(x)$ over the distribution of the relevant population.

Among the methods under consideration, The Bayesian Adaptive Regression Trees (BART) is the most straightforward conceptually. It involves fitting a BART model for $\mu_z(x) = \mathbb{E}[Y \mid X = x, Z = z]$ predicting $\mu_1(x)$ and $\mu_0(x)$ with the fitted model and then estimating the treatment effect as the difference between the predicted conditional means for each treatment option. BART works by fitting a sum of many small regression trees to model the outcome flexibly, then estimating treatment effects as differences in predicted outcomes under each treatment ([Chipman et al., 2010](#); [Hill, 2011](#)).

The second approaches, Neural networks estimate causal effects by learning complex, non-linear relationships between covariates and outcomes. They do this through layers of interconnected nodes, which transform input features into derived representations that capture hidden patterns. These representations help predict counterfactual outcomes in different treatment scenarios ([McCulloch and Pitts, 1943](#)).

The fourth method, Causal Trees, focuses on estimating heterogeneous treatment effects by partitioning the covariate space into subgroups (leaf nodes) where treatment effects are relatively homogeneous. The tree is grown using a splitting criterion specifically designed to maximize differences in treatment effects between branches rather than minimizing prediction error. Once the tree is built, the treatment effect for a given covariate profile x is estimated as the difference in average outcomes between treated and control units within the corresponding leaf node:

$$\hat{\tau}(x) = \frac{1}{|L_1(x)|} \sum_{i \in L_1(x)} Y_i - \frac{1}{|L_0(x)|} \sum_{i \in L_0(x)} Y_i,$$

where $L_1(x)$ and $L_0(x)$ denote the sets of treated and control units within the leaf node associated with covariate profile x ([Athey and Imbens, 2016](#)).

The last two approaches, the causal forests and the causal boosting, address treatment effect

heterogeneity more explicitly by employing modified regression trees as building blocks for ensemble models for $\tau(x)$. Regression trees are fitted using modified splitting criteria designed to maximize heterogeneity in treatment effects. Treatment effects are then estimated based on the samples from both the treated and control groups within each leaf node (Athey et al., 2019; Powers et al., 2018).

The benchmark method, Inverse Probability Weighting (IPW), is used to estimate causal effects in observational studies by creating a pseudo-population in which treatment assignment is independent of observed covariates. Each individual is weighted by the inverse of the probability of receiving the treatment they actually received, given their covariates. Formally, the weights are defined as

$$w_i = \frac{Z_i}{e(X_i)} + \frac{1 - Z_i}{1 - e(X_i)},$$

where Z_i is the treatment indicator and $e(X_i) = P(Z_i = 1 | X_i)$ is the propensity score. Applying these weights allows unbiased estimation of average treatment effects by balancing the distribution of covariates between treated and control groups (Powers et al., 2018).

We will assess the model’s performance by examining the root mean square error of the estimates of individual-specific CATE, $\hat{\tau}_i(x)$, which can be written as

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{\tau}_i(x) - \tau_i(x))^2}.$$

We will additionally present the performance of the ATE estimate (marginal risk difference) in the dataset. This is quantified by the absolute relative bias in the ATE estimate, expressed as $100 \times \frac{|\hat{\tau} - \tau|}{\tau}$, where $\hat{\tau}$ represents the estimated ATE, and τ is the true ATE in the dataset. In this study, inverse probability weighting (IPW) was adopted as a benchmark approach for evaluating performance.

5 Simulation experiments

The data generating procedure in this study is empirical data based where empirical covariates and treatment assignment data are directly used and only outcomes are simulated based on non-parametric models of the outcomes. Using the simulated dataset, we will estimate conditional average treatment effects (CATEs) with the proposed methods and evaluate how well each method performs.

5.1 Empirical data

The Bangladesh Demographic and Health Survey (BDHS) 2022 data will be used for this study. The outcome variable is antenatal care (ANC) utilization, and the treatment of interest is maternal

higher education. Both the outcome and treatment variables are binary. Maternal higher education will be defined based on the woman’s highest level of educational attainment, with women who have completed secondary or higher education classified as having higher education and those with lower educational attainment classified otherwise. ANC utilization will be defined based on the number of antenatal care visits received during the most recent pregnancy. Women who received at least four ANC visits will be classified as having adequate ANC utilization ($ANC = 1$), while those who received fewer than four visits will be classified as having inadequate ANC utilization ($ANC = 0$), consistent with the conventional WHO recommendation (BDHS, 2022; World Health Organization, 2016).

5.2 Data generating process

The data-generating process in this study follows the framework proposed by (Wendling et al., 2018). To estimate the average effect of a binary treatment $Z \in 0, 1$ on a binary outcome $Y \in 0, 1$, we define the conditional risk difference as the estimand for the conditional average treatment effect (CATE). The treatment effect function is defined as

$$\tau(x) = \pi^1(x) - \pi^0(x),$$

Here, $\pi^1(x)$ and $\pi^0(x)$ represent the probabilities of the potential outcomes under treatment and control, respectively, and are defined as,

$$\pi^1(x) = E[Y = 1|X = x, Z = 1],$$

$$\pi^0(x) = E[Y = 0|X = x, Z = 0].$$

We design a simulation framework to produce cross-sectional observational data while introducing only a minimal number of synthetic variables. In particular, the covariates X and treatment indicators Z are drawn from the observed data, whereas only the binary factual outcome $Y_{\text{sim}} \in \{0, 1\}$ and the treatment effect function $\tau(x)$ are generated synthetically as specified below.

1. The conditional outcome probability $\pi^z(x)$ is estimated using nonparametric methods to avoid imposing specific functional form assumptions. The fitted model is used to predict the conditional probabilities $\pi^1(x)$ and $\pi^0(x)$ given control and treatment, respectively.
2. These two potential outcomes are then employed to construct the treatment effect function on the linear predictor scale within a logistic regression framework.

$$\gamma(x) = \text{logit}[\pi^1(x)] - \text{logit}[\pi^0(x)] + c,$$

Here, c is a constant that is determined iteratively to ensure that the mean of $\gamma(x)$ matches the logarithm of the odds or hazard ratio estimated from the original observational study.

3. The probability of the potential outcome under treatment is recalculated as

$$\pi_{\text{adjusted}}^1(x) = \text{logit}^{-1}[\text{logit}[\pi^0(x)] + \gamma(x)],$$

and the treatment effect, expressed on the probability scale as the risk difference, is calculated using

$$\tau(x) = \pi_{\text{adjusted}}^1(x) - \pi^0(x).$$

4. The factual outcome is drawn as

$$Y \sim \text{Bernoulli}(\pi_{\text{factual}}(x)), \text{ with } \pi_{\text{adjusted}} = \pi_{\text{adjusted}}^1(x)Z + \pi^0(x)(1 - Z).$$

The outcome model with treatment effects follows a standard logistic regression framework (Step 3), but the functional forms of the marginal effects and the interaction between baseline covariates and treatment status are left unspecified. Through Steps 2 and 3 of the data-generating process, we set the mean of $\tau(x)$ to a desired value, such as one based on marginal effect estimates from the literature. Because marginal effects are commonly expressed as odds or hazard ratios, we instead set the mean of the function $\gamma(x)$ (Step 2) to the logarithm of the estimated odds or hazard ratio. We employed a deep feedforward neural network to flexibly estimate $\pi^1(x)$, while deliberately avoiding models similar to those used for estimating the CATEs. Each simulated dataset included X , Z , Y_{sim} , $\pi_{\text{factual}}(x)$, and $\tau(x)$, which were saved for subsequent performance evaluation. The generated data are used to evaluate the best performing model among those described in the methodology section.

5.3 Sensitivity analysis

In this study, we assume the presence of a binary unmeasured confounder that jointly affects maternal early marriage and child stunting. We aim to examine whether our estimator is robust under different assumptions about the unmeasured confounder. If the unmeasured confounder is collinear with a measured confounder, then adjusting for the unmeasured confounder would not substantially affect the estimated causal association (Groenwold et al., 2010). Therefore, we assume that the unmeasured confounder is not collinear with the measured confounders. The simulation design follows the approach proposed by (Groenwold et al., 2010) :

This method is based on specifying the strength of association between the unmeasured confounder and the treatment, the association between the unmeasured confounder and the outcome, and the prevalence of the unmeasured confounder. Let Y denote the outcome variable, Z the treatment variable, and C a dichotomous unmeasured confounder. Also let m_{ijk} denote the sample count for which $Z = i$, $Y = j$ and $C = k$. To quantify the associations, we consider the risk difference (RD)

for $C - Z$ and $C - Y|Z$. We generate C using a binomial distribution with size determined by the cross-tabulation and equal probability. The conditional association between C and Y given Z is shown below :

Table 1: The association between a binary unmeasured confounder (C) and a binary outcome (Y), conditional on strata of the binary treatment (Z)

C	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$C = 0$	m_{000}	m_{010}	m_{100}	m_{110}
$C = 1$	m_{001}	m_{011}	m_{101}	m_{111}

The marginal association between Z and Y is shown below :

Table 2: The marginal association of the binary unmeasured confounder (C) with the binary treatment measure (Z)

C	Z		Total
	$Z = 0$	$Z = 1$	
$C = 0$	a_{00}	a_{01}	N_{Z0}
$C = 1$	a_{10}	a_{11}	N_{Z1}
Total	N_{Z0}	N_{Z1}	N

To calculate these cell counts of 2×2 table , we use the following formulas:

$$RD_{CZ} = P(Z = 1|C = 1) - P(Z = 1|C = 0),$$

$$a_{11} = N[RD_{CZ}P_C(1 - P_C) + P_ZP_C],$$

$$a_{10} = N[P_C + RD_{CZ}P_C(1 - P_C) - P_ZP_C],$$

$$a_{01} = N[P_Z(1 - P_C) - RD_{CZ}P_C(1 - P_C),]$$

$$a_{00} = N[(1 - P_C) - P_Z(1 - P_C) + RD_{CZ}P_C(1 - P_C)],$$

where, $P_Z = \frac{N_{Z1}}{N}$ represents the proportion of individuals exposed in the study population, P_C

is the prevalence of confounder, and RD_{CZ} is the Risk difference (RD) between treatment and counfounder.

We assume that the conditional association between C and Y given Z is constant across levels of Z; that is, no interaction occurs.

To calculate these cell counts of three level table , we use the following techniques:

$$P_{C|Z=1} = \frac{a_{11}}{a_{01}+a_{11}} = \frac{RD_{CZ}P_C(1-P_C)+P_ZP_C}{P_Z},$$

$$RD_{CY|Z=0} = \frac{m_{011}}{m_{001}+m_{011}} - \frac{m_{010}}{m_{000}+m_{010}},$$

$$RD_{CY|Z=1} = \frac{m_{111}}{m_{101}+m_{111}} - \frac{m_{110}}{m_{100}+m_{110}},$$

$$P_1 = \frac{m_{111}}{m_{111}+m_{110}} = \frac{RD_{CY|Z=1}(1-P_{C|Z=1})+P_{Y|Z=1}P_{C|Z=1}}{P_{Y|Z=1}}.$$

We generate multiple datasets by varying the prevalence of the unmeasured confounder, the magnitude of the risk difference between the unmeasured confounder and the treatment, and the magnitude of the risk difference between the unmeasured confounder and the outcome conditional on treatment. These simulated datasets are used to evaluate the robustness of both the selected model and the benchmark method.

6 Time Frame

Table 3: Time frame of the study

Time	Task
January 2026	Idea formulation and literature review
February 2026	Preparation and presentation of the thesis proposal
June 2026	Methodology development
July 2026	Coding
August 2026	Computation and simulation
September 2026	Completion of the first draft of the thesis report
October 2026	Final draft completion, submission, and thesis defense

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